Sai Krishna Chaitanya Annavazala

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Education

New York University — New York, United States
Master of Science in Computer Engineering — *Jan 2022 – Dec 2023*

Jawaharlal Nehru Technological University, Hyderabad — Hyderabad, India
Bachelor of Technology in Electrical and Electronics Engineering — *Jul 2017– Jul 2021*

Technical Skills

Programming Languages & Frameworks: Python, Shell Scripting, C/C++, JavaScript
Databases & Web Technologies: MySQL, HTML, CSS
Machine Learning: TensorFlow, Keras, Model Deployment
Tools & Platforms: Git, JIRA, Selenium, REST API/RESTful Services, SDLC
Containerization & Workflow Management: Docker, Apptainer

Experience

Data Analyst C — Philadelphia, United States
University of Pennsylvania — DiCIPHR Lab — *Nov 2024 – Present*

- Designed and containerized an end-to-end MRI processing pipeline using **Apptainer**, integrating DTI preprocessing, brain registration, ROI statistics extraction, and structural connectivity analysis for reproducible HPC deployment.
- Processed and standardized MRI data for **3,500+ subjects** across six multi-site neuroimaging cohorts using containerized pipelines, including **PING, PPMI, TBIRI, TRACK-TBI Phases 1 and 2, and TBIRI-LATMI**, enabling reproducible large-scale analysis across heterogeneous datasets.
- Developed **DenseNet**-based deep learning models to automatically classify DTI and T1-weighted MRI scans as artifact-present or artifact-free, improving the efficiency and reliability of image quality control workflows.
- Contributed to training and evaluating **PIGMENT**, a deep learning framework for segmenting markers of axonal injury in immunohistochemistry images.
- Implemented **UniverSeg** and **PathoSAM** as comparative baselines and evaluated their segmentation performance against **PIGMENT**, and contributed to writing the manuscript.
- Maintained lab's website using **HTML, CSS, and JavaScript**: diciphrlab.com.

Bioinformatics Analyst — New York, United States
NYU Grossman School of Medicine — *Jan 2024 - Jul 2024*

- Implemented bioinformatics data pipelines streamlining high-throughput sequencing.
- Conducted data analysis using Python, identifying key genetic variants and patterns in large genomic datasets.
- Developed custom scripts and visualization tools to create insightful plots and graphs, enhancing data interpretation.

High Content Image Analysis Research Associate — Rockville, United States
Axle Informatics — *Sept 2023 - Dec 2023*

- Implemented an automated plasma cell identification pipeline using U-Net for nucleus segmentation and DeepLabv3+ for cytoplasm segmentation in microscopic images.
- Designed and executed comprehensive test plans and validation protocols for high-content image analysis software.
- Collaborated with cross-functional teams to improve software usability and drive measurable gains in product quality.

Graduate Employee Adjunct — New York, United States
New York University Courant Institute of Mathematical Sciences — *Jan 2023 – May 2023*

- Led recitation sessions for Computer System Organization, teaching software quality principles and C++ best practices under Professor Gizem Kayar, while mentoring students in debugging, code optimization, and structured problem-solving.
- Designed recitation worksheets and hands-on exercises to reinforce software testing methodologies, improving student comprehension and practical application.

Programmer Analyst — Chennai, India
Cognizant Technology Solutions — *Mar 2021 – Dec 2021*

- Analyzed and designed test plans for complex software systems, ensuring thorough validation and verification of court docket and document processing on LexisNexis Courtlink.
- Conducted manual testing using JIRA and automated testing with Selenium, adhering to strict quality assurance protocols to meet industry standards.

Publications, Presentations, and Research Contributions

- **Sai Krishna C. Annavazala**, Nima Broomand Lomer, Tyler Spears, Drew Parker, Ragini Verma. *"NeuroQC-Assist: AI-Assisted Neuroimaging Quality Control."* Accepted for presentation at the ***Conference on Artificial Intelligence in Medical Imaging (SIIM-CAIMI)***, Philadelphia, PA, October 2026.
- Pushkar Ambastha, Javid Dadashkarimi, **Sai Krishna C. Annavazala**, Drew Parker. *"PIGMENT: A deep learning framework for Porcine Immunohistochemistry seGMENTation"* Manuscript under review at ***AI in Neuroscience***
- Rui Sherry Shen, Livia Rodrigues, Yusuf Osmanlioglu, Drew Parker, **Sai Krishna C. Annavazala**. *"Parsing Altered Brain Connectivity in Neurodevelopmental Disorders by Integrating Graph-Based Normative Modeling and Deep Generative Networks."* ***Communications Medicine (Nature)***
- Dadashkarimi J., Parker D., Broomand Lomer N., et al. *"PIGSKIN: PIG SKull Stripping with Synthetic Images and Neural Networks."* ***AI in Neuroscience*** **Acknowledged contributor**: created and maintained the containerized deployment environment for PIGSKIN.